Sets 2 — Practical problems · Three sets · Shading · Set-builder notation

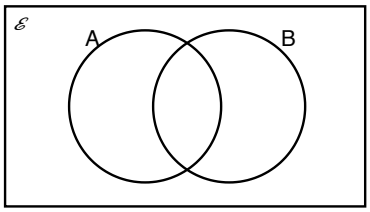
Marks in brackets. ★ = harder question. $\mathscr{E}$ is the universal set; A' is the complement of A.

1 In a class of 30 students, 18 study French, 15 study Spanish and 5 study neither language. [5]

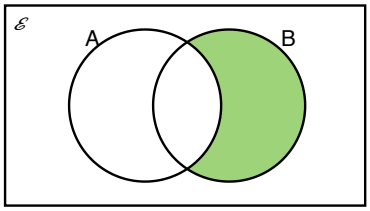
a Draw a Venn diagram, using x for the number who study both. **b** Find x.

c How many students study exactly one of the two languages?

2 On two copies of the diagram, shade the sets **a** $A \cap B'$ and **b** $(A \cup B)'$. [4]



3 Describe the shaded set using set notation. [3]



4 $C = \{x : \ -3 \leq x < 3,\ x \in \mathbb{Z}\}$. List the members of C and write down $n(C)$. [3]

5 Write in set-builder notation the set of positive multiples of 7. [2]

6 In a sports club of 120 members, 50 play basketball, 60 play volleyball and 45 play handball. 15 play basketball and volleyball, 20 play volleyball and handball, 12 play basketball and handball, and 6 members play none of the three sports. Use a Venn diagram to find how many members [8]

a play all three sports **b** play only volleyball.

Total: 25 marks

Check: put x in the overlap and work outwards, then add every region to the total · "neither" goes outside the circles · $A \cap B'$ = in A but not B; $(A \cup B)'$ = outside both · $\mathbb{N}$ natural numbers, $\mathbb{Z}$ integers, $\mathbb{Q}$ rationals, $\mathbb{R}$ reals · $\{x : \text{condition}, x \in \text{set}\}$ reads "the set of x such that…".